MILITARY SPECIFICATION

CONNECTION, FLEXIBLE; PIPE LOOPS, FOR STEAM TO MAIN AND AUXILIARY TURBINES (SIZES 4 AND 6 INCHES IPS)

1. SCOPE

1.1 Scope.- This specification covers flexible pipe loops for installation in steam piping systems as connections to resiliently-mounted main and auxiliary turbines for the purpose of absorbing high frequency vibrations and occasional large motions due to shock and drift of the resilient-mounts.

1.2 Classification.- Loops shall be of the following types and sizes, as specified (see 6.2):

Types:
- Type I - Port
- Type II - Starboard

Sizes:
- 4 inches, iron pipe size (i.p.s.)
- 6 inches, i.p.s.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
- PPP-B-601 - Boxes, Wood, Cleated - Plywood.
- PPP-B-621 - Boxes, Wood, Nailed and Lock - Corner.

MILITARY
- JAN-P-100 - Packaging and Packing for Overseas Shipment, General Specification.
- MIL-P-116 - Preservation, Methods of.
- MIL-L-10547 - Liners, Case, and Sheet, Overwrap, Water - Vaporproof or Waterproof, Flexible.
- MIL-C-16173 - Corrosion Preventive Compound, Solvent Cutback, Cold Application.
- MIL-T-20157 - Tube and Pipe, Carbon Steel, Seamless.

STANDARDS

MILITARY
- MIL-STD-129 - Marking for Shipment and Storage.
- MIL-STD-271 - Nondestructive Testing Requirements for Metals.
- MIL-STD-278 - Welding and Allied Processes for Machinery for Ships of the United States Navy.

DRAWINGS

BUREAU OF SHIPS
- 810 - 1385937 - Piping, Flexible- 4 Inch Steam Loop for Auxiliary Turbines.
- 810 - 1385938 - Piping, Flexible- 6 Inch Steam Loop for Main Propulsion Turbines.

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PUBLICATIONS

BUREAU OF SHIPS
NAVSHIPS 250-582 - Fabrication and Control of Steel Pipe and Tubing for Welded Piping Systems.

(Copies of specifications, standards, drawings and publications required by contractors in connection w. specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.- The material of the loops shall be carbon steel pipe in accordance with type E of MIL-T-20157, procured in the normalized condition. Pipe shall be 4 or 6 inch i.p.s. as specified (see 1.2) with schedule 40 nominal thickness (0.237 inch for 4 inch i.p.s. and 0.280 inch for 6 inch i.p.s.) (see table I). Pipe shall not be used for loops unless it has passed the ultrasonic tests in accordance with 4.3.

3.1.1 The material covered herein is for use in steam piping systems as connections to resiliently-mounted main and auxiliary turbines on submarines and for this reason operates under critical conditions. The application is such that the use of incorrect or defective materials may result in serious personal injury, loss of life or loss of vital shipboard systems. For this reason, this material has been designated as "Target Material" and special material control procedures are required to insure to the maximum extent possible that correct material is used (see 4.2).

3.2 Construction.-

3.2.1 Fabrication procedures.- Fabrication procedures shall be in accordance with the applicable requirements of NAVSHIPS 250-582. The configuration of the pipe loops shall be in accordance with Drawing 810-1385937 for the 4 inch size and Drawing 810-1385938 for the 6 inch size. Either hot or cold forming may be employed. Tolerances shall be in accordance with table I.

Table I - Tolerances

Pipe sizes	Maximum allowable out-of-roundness	Minimum wall thickness at any point after fabrication	Maximum allowable contour variation
Inches	Inch	Inch	Inch
4	0.255	0.196	1/2
6	.331	.230	1/2

Pipe loops shall be free of wrinkles. Removal of wrinkles by machining or grinding is not permitted.

3.2.2 Welding.- All welding shall be in accordance with MIL-STD-278. Pipe butt welds shall be made by the inert gas, tungsten arc process using consumable inserts.

3.2.3 Location of weld joints.- Weld joints in the pipe loops need not be located as shown on Drawing 810-1385937 or Drawing 810-1385938, as applicable. A lesser number of weld joints is preferable. However, joints shall not occur in any of the bends or the bend tangent areas.

3.2.4 Stress relieving.- All pipe loops shall be stress relieved, in whole or in part after forming and welding in accordance with NAVSHIPS 250-582. Stress relief temperature shall be 1150°F ± 25°F. The assembly shall be adequately supported during the stress relief operation to prevent distortion.

3.3 Material marking and identification.-

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3.3.1 Marking.- Both ends of each loop shall be legibly and permanently marked by means of electro-etch or by separate metal tag tack-welded to the pipe. This marking shall be within 2 inches of either end. Marking shall include the following:

(a) Coding of the name or trade-mark of the pipe manufacturer and loop manufacturer.
(b) Heat number or heat code.
(c) MIL-T-20157/TYPE E/SCH 40.

3.3.2 Identification.- Identification shall be made on steel tags wired to the middle of the assembly by steel wire. Tags shall show the following:

(a) Contract number.
(b) Drawing number.
(c) Port or starboard loop.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Material certification.- The contractor shall certify in writing that the chemical analysis and physical properties of the material used in the construction of the loops furnished conform to the material requirements specified herein. Such certification shall contain actual test results, examination or other verifiable quality data.

4.3 Examination and tests.- Examination and tests shall be performed on all pipes and pipe loops, as applicable. Examination and tests shall be as specified in 4.3.1 through 4.3.4.2.

4.3.1 Ultrasonic inspection (pipe).- Prior to fabrication, all pipe shall be ultrasonically inspected in accordance with MIL-STD-271. Equipment shall be calibrated to 3 percent sensitivity as specified for radioactive systems, and any piping having a defect in excess of the above shall not be used.

4.3.2 Radiographic inspection (welded joints).- All welded joints shall be radiographically inspected in accordance with MIL-STD-278.

4.3.3 Examination (pipe and loops).- Surface of the loops shall be free of cracks, linear indications, die stamping and punch marks. If any indications of these types are found, the loop shall be ground smooth and the loop shall be considered satisfactory provided the wall thickness is not less than that specified in table I. The loop shall also be thoroughly examined to assure that out of roundness and contour variation is not above that specified in table I. The bent area of the pipe shall be vidigaged along the entire length of the extrados and at any ground areas, as well as any suspect areas to assure that wall thickness is not less than that specified in table I.

4.3.4 Tests.-

4.3.4.1 Hydrostatic tests (loops).- Each loop shall be subjected to a hydrostatic test pressure of 1275 p.s.i. Any loops which fail to pass the test shall be rejected.

4.3.4.2 Steam pressure test.- The loops shall be able to pass a steam pressure test of 850 psi with a minimum temperature of 530°F. The contractor is not required to perform this test.

5. PREPARATION FOR DELIVERY

5.1 Cleaning, preservation and packaging.- The loops shall be free of all loose scale and foreign matter and shall be cleaned by abrasive blast or acid pickling (including rinsing or other means) or by a combination of both and shall be packaged in accordance with method III of MIL-P-116 using preservative grade 4 of MIL-C-16173.

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5.2 Packing.- Loops shall be packed in accordance with PPP-B-601 or PPP-B-621. Anchoring, blocking, bracing and cushioning to prevent damage or undue strain from handling, shipment and storage shall be in accordance with JAN-P-100. Caseliners shall be in accordance with MIL-L-10547.

5.3 Marking.- In addition to any special marking required, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- Loops are intended for use in steam piping systems in submarines as inlet connections to resilient-mounted main and auxiliary turbines for the purpose of absorbing high frequency vibrations, occasional large motions lue to shock and drift of the resilient-mounts.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and size required (see 1.2).
(c) Number required.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Ships
(Project 4730-N053Sh)

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

BUDGET BU. NO. 45-R309

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification.)* PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY *(Print name and activity)*	DATE

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